



ARC RESEARCH · SPECIAL TOPICS

Money for machines

Crypto as the payment layer for AI agents: paying for APIs, data and compute

Rails · Use cases · Data · Markets · Regulation

23 September 2026 · Ashley Castelino · Educational purposes only, not investment advice

EXECUTIVE SUMMARY

The rails are built; the agents have not shown up yet

Stablecoins fit agent payments. The rails exist; the demand doesn't yet.

\$312bn

Stablecoin supply

DeFiLlama

8.7M

x402 transfers in one week

Only \$0.37M moved

0.6–7.5%

x402 value from AI agents

TRM Labs

49–50

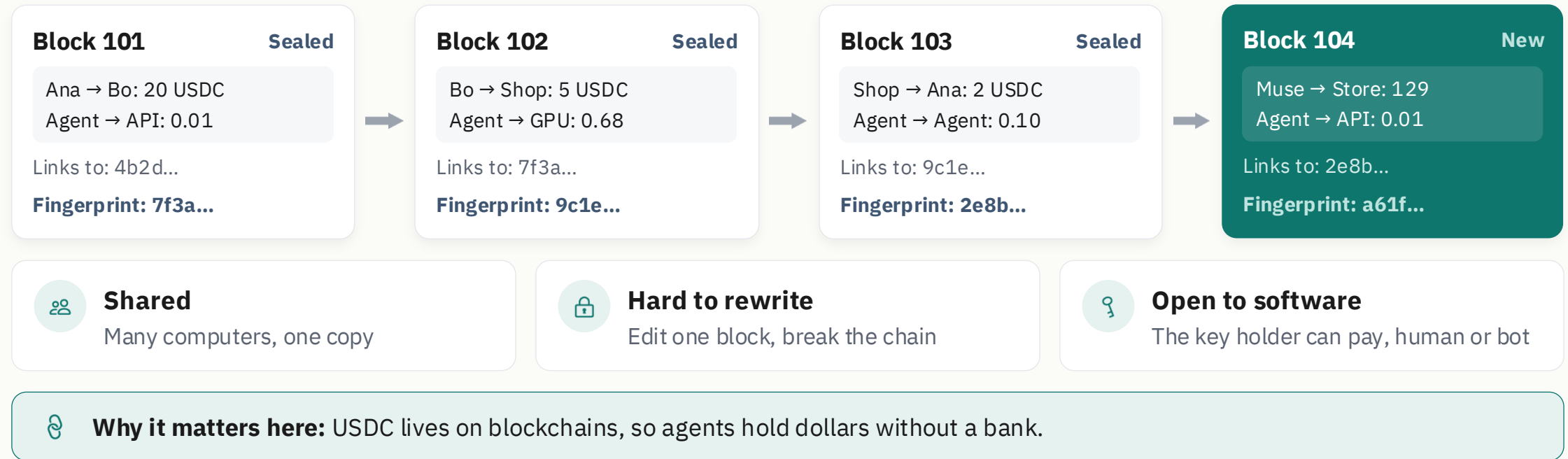
**Senate vote stalled
CLARITY**

GENIUS Act is law

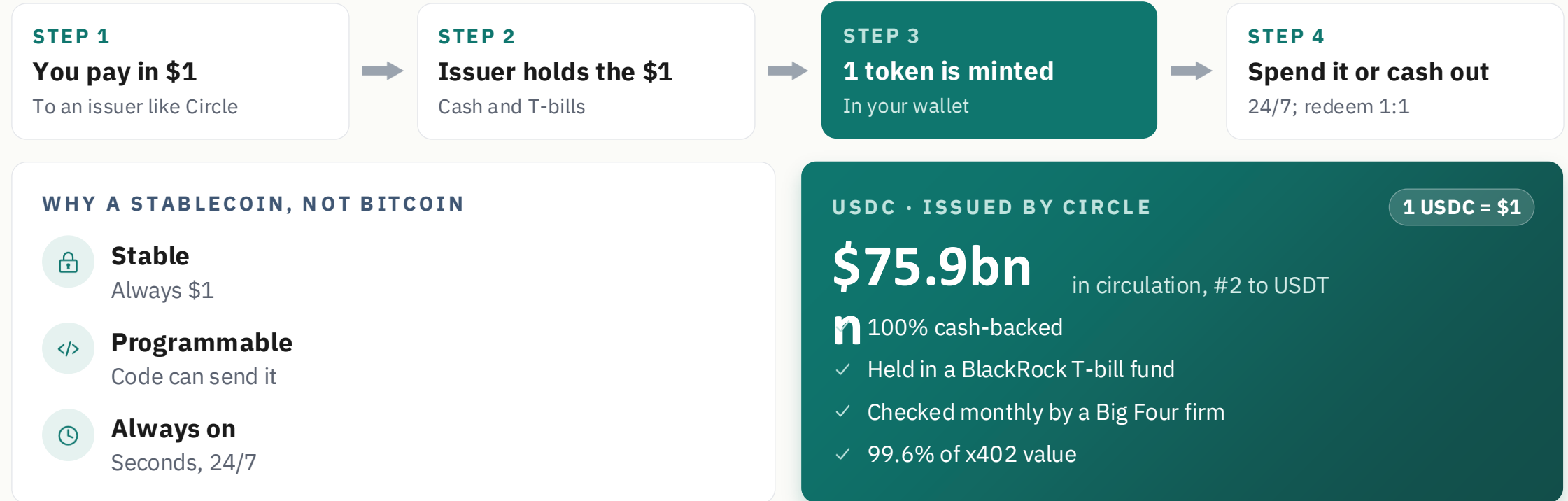


A blockchain is a shared record no one can quietly edit

ILLUSTRATIVE



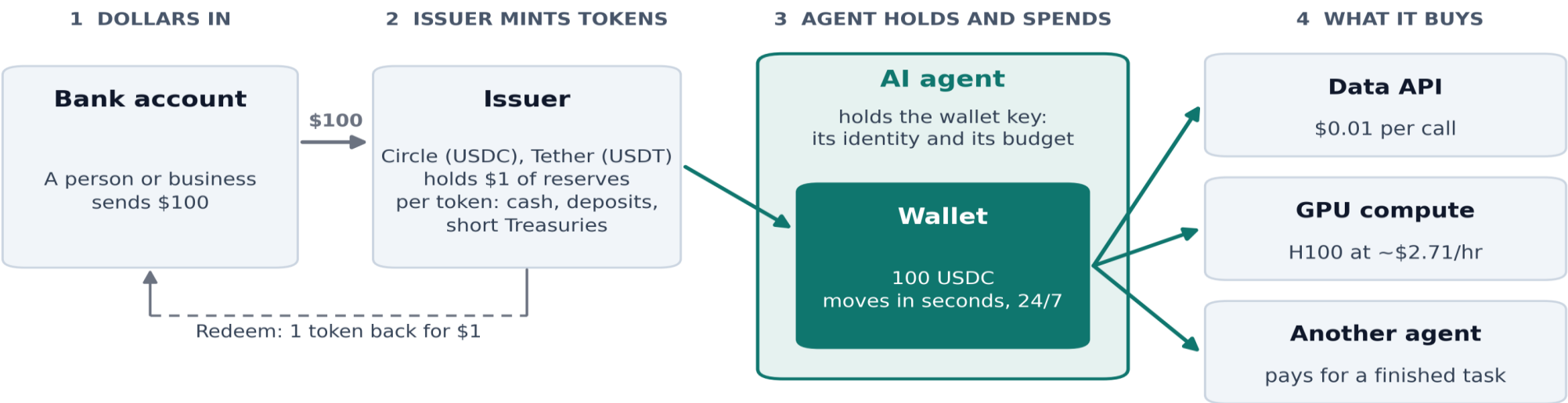
A stablecoin is a digital dollar on a blockchain



Sources: Circle, DeFiLlama, TRM Labs

A stablecoin is a dollar that software can hold and spend

A dollar goes in at the issuer, comes out as a token in a wallet, and an AI agent holding the key can spend it on its own



\$312bn in stablecoins today | USDC \$76bn | 1:1 reserves required by US law

Paying by card

- Needs a person with ID to open the account
- 30c minimum fee: 3,003% of a 1-cent payment
- Settles in days, on business hours

Paying in stablecoins

- Needs only a wallet key, which software can hold
- \$0.001 to settle: 10% of a 1-cent payment
- Settles in seconds, any hour

Source: supply DeFiLlama, 23 Sep 2026 (total \$312bn; USDT \$183bn, USDC \$76bn); GENIUS Act, signed 18 Jul 2025 (1:1 reserves in cash, insured deposits, short Treasuries); CoinGecko x402 API endpoint \$0.01 per request (Coinbase Developer Platform); Akash H100 marketplace average \$2.71/hr (akash.network, 12 Sep 2026); Stripe standard card pricing 2.9% + 30c; Coinbase CDP x402 facilitator \$0.001 per settlement. Schematic, our illustration: simplified; redemption is typically available to the issuer's direct customers; agent purchases shown are examples.

ARC RESEARCH | CRYPTO X AGENTS | THE DESIGN ARGUMENT

Crypto's rough edges for people are features for software

The traits that kept crypto out of everyday human payments are the ones an autonomous agent needs

CRYPTO TRAIT	FOR A PERSON: FRICTION	FOR AN AGENT: FIT
Private keys, self-custody	Lose the seed phrase, lose it all	Stores a key like any credential; signs in milliseconds
Irreversible payments	No chargebacks, so scams stick	Deterministic settlement; limits set in code, not after the fact
42-character addresses	One typo sends the funds astray	Copies strings exactly, every time
A fee on every transfer	An irritant on a coffee	About \$0.001: workable across millions of micro-calls
No bank account to open	Most people already have one	Cannot pass KYC; a wallet is the only account it can get
Always on, API-native	Payday once a fortnight is fine	Runs 24/7 and pays mid-request over HTTP 402
Programmable money	Needs an app to make sense of it	Reads the contract and enforces budgets directly

Source: our framework. Fee: Coinbase CDP x402 facilitator, \$0.001 per settlement after 1,000 free per month (docs.cdp.coinbase.com, 23 Sep 2026). Agents and bank accounts: Google Cloud's Richard Widmann at Consensus Miami (CoinDesk, 10 May 2026); OpenLedger's Ram Kumar (The Block, 20 Aug 2026). HTTP 402 flow: x402 specification. BlackRock's 'The Machine-Native Economy' (Sep 2026) makes a similar case for machine-to-machine payments. Caveat: Bitcoin was designed in 2008 as peer-to-peer cash for people; this is an argument about fit, not origin.

Crypto fixes most agent-payment worries, but not the undo button

THE WORRY	WHAT CRYPTO DOES	STILL OPEN	
 Handing an agent your card	Own wallet, only what you fund	Stolen key, lost balance	Solves
 Runaway spending	Hard caps in the wallet	Caps can't judge a bad buy	Solves
 Tiny payments cost too much	~\$0.001 per payment	Little real volume yet	Solves
 Agents can't open accounts	No bank account needed	Liability untested	Partly
 Proving who paid, and why	Public, timestamped ledger	Spending is public	Partly
 Mistakes and fraud	No chargebacks	Cards win here	Worse

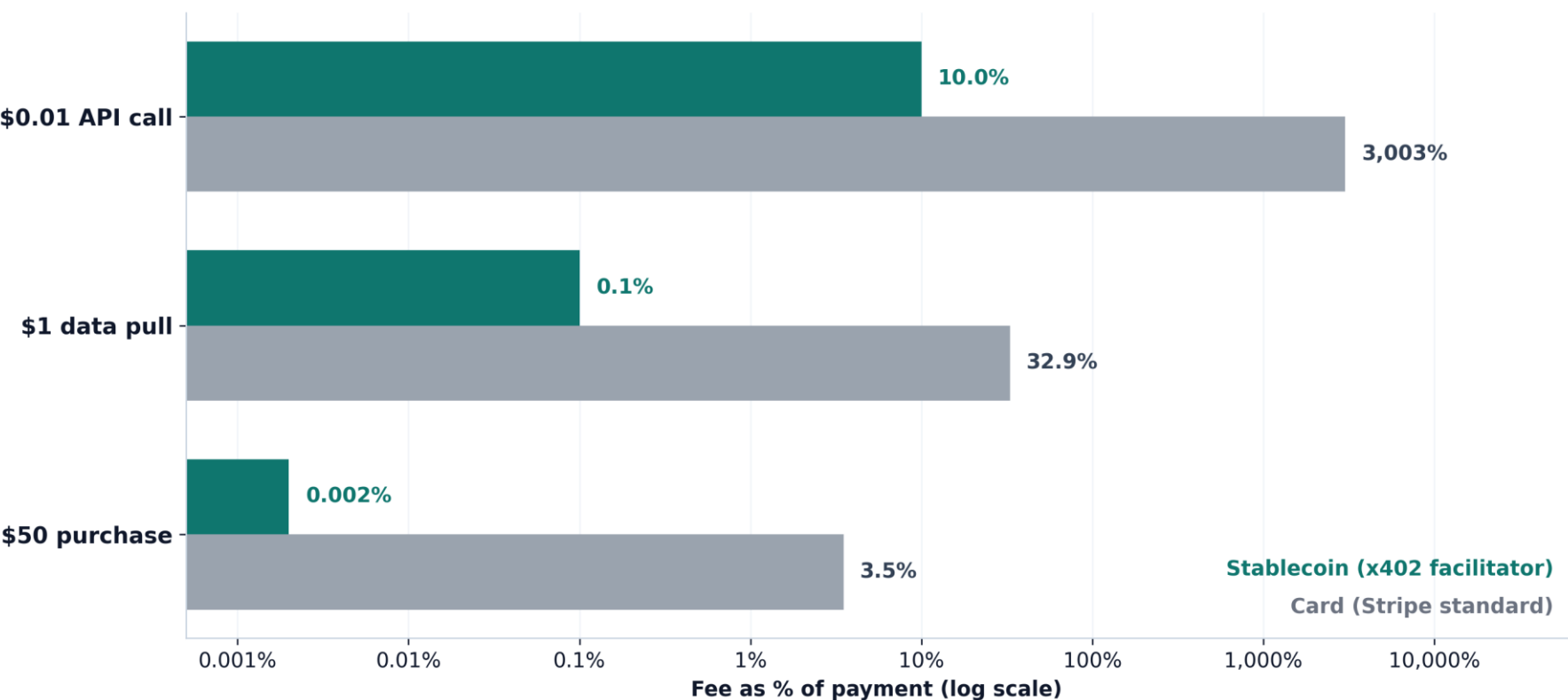


01 · WHY AGENTS NEED DIFFERENT RAILS

ARC RESEARCH | CRYPTO X AGENTS | WHY THE RAILS MATTER

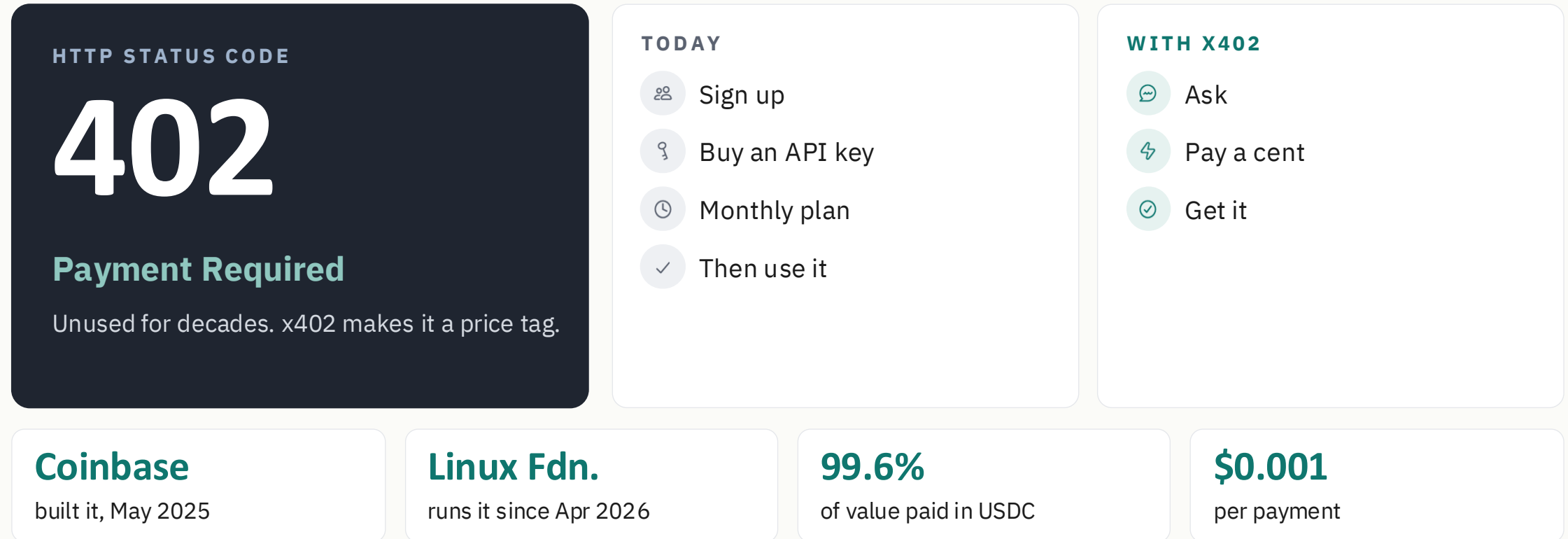
A fixed card fee swamps a one-cent payment; a stablecoin settlement fee does not

Fee as % of payment value: Stripe standard card pricing vs Coinbase's x402 facilitator fee (log scale)



Source: Stripe US standard online card pricing, 2.9% + 30c per successful domestic transaction (stripe.com/pricing, checked 23 Sep 2026).
Coinbase CDP x402 facilitator, \$0.001 per settlement after 1,000 free per month (docs.cdp.coinbase.com, checked 23 Sep 2026). Derived: fee % = fee / payment x 100, our calculation.
Excludes interchange variation, FX, network gas not covered by the facilitator, and wallet on/off-ramp costs. Illustrative payment sizes.

x402: the web's missing checkout for machines



Sources: Coinbase docs; Linux Foundation; TRM Labs

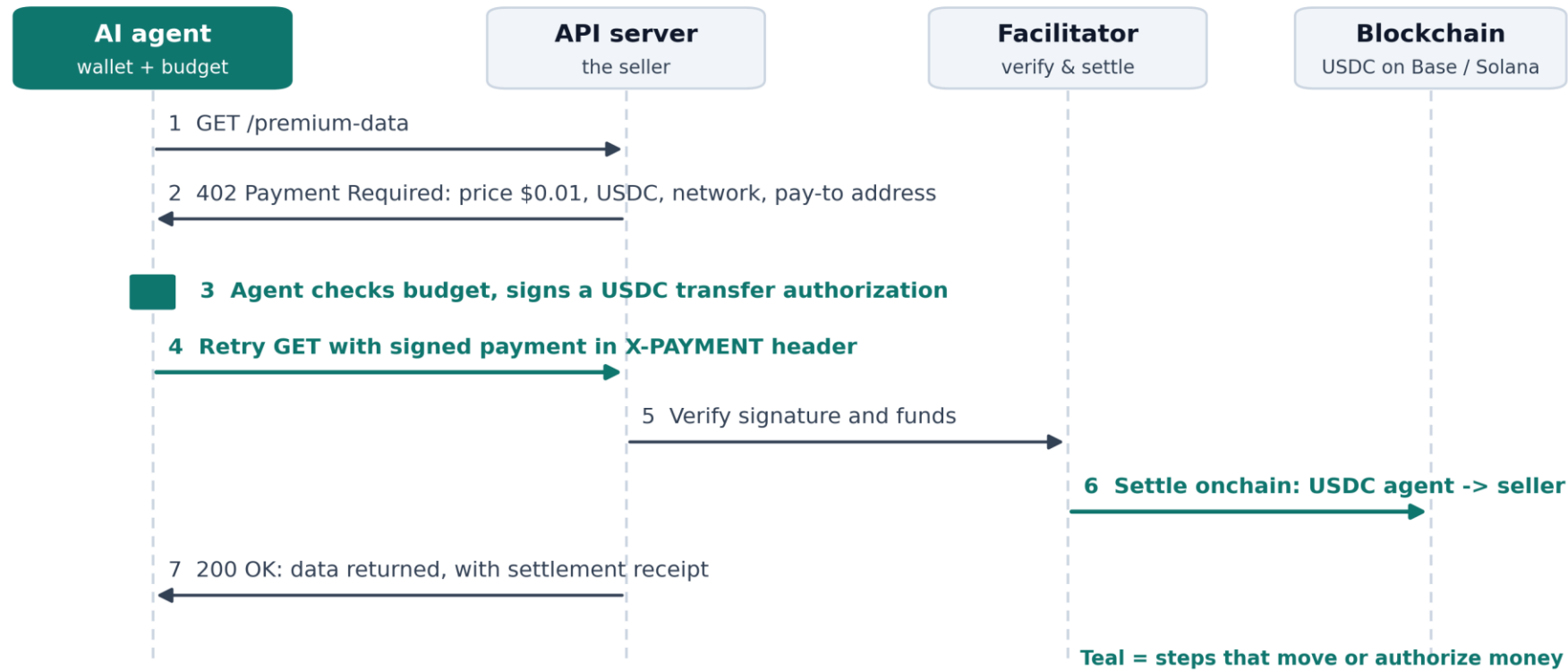


01 · WHY AGENTS NEED DIFFERENT RAILS

ARC RESEARCH | CRYPTO X AGENTS | HOW AN X402 PAYMENT WORKS

One HTTP round trip: the server quotes a price, the agent signs, USDC settles onchain

No account, API key or card on file; the agent's wallet is its identity and its budget



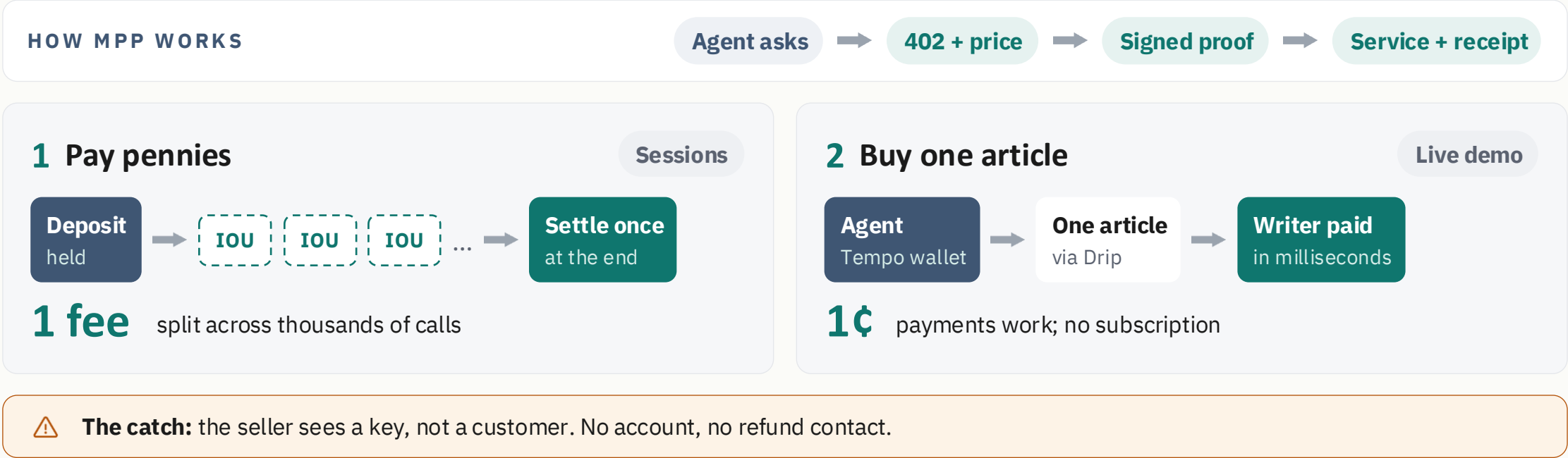
Source: x402 protocol specification and Coinbase CDP documentation (docs.cdp.coinbase.com/x402), checked 23 Sep 2026; x402 Foundation under the Linux Foundation since 2 Apr 2026. Schematic, our illustration: simplified; facilitator may be self-hosted; headers and payload fields abbreviated. Settlement shown for USDC on Base or Solana, the most-used networks.

Four standards, two camps, converging

Standard	Backer	What it does	Settles in
x402	Coinbase → Linux Foundation	Pay per request	USDC
Machine Payments Protocol	Stripe + Tempo	Micro and recurring bills	Stablecoins, cards, BNPL
AP2	Google → FIDO Alliance	Proves what you approved	Cards, stablecoins
Visa TAP · Mastercard Agent Pay	Visa, Mastercard	Verify agents at checkout	Card tokens

Not rivals: Visa, Mastercard, Stripe and Google all joined the x402 Foundation.

Stripe's MPP: two ways agents pay



Coinbase's pitch: six things agents will pay for

APIs and data

Live

A tool call mid-task

CoinGecko: \$0.01 a call

AI model access

Live

No subscription

BlockRun, per call

Compute and storage

Early

Rent GPUs, no sign-up

io.net Agent Cloud

Publisher paywalls

Launched

Charge bots, don't block

AWS, Cloudflare

Agent-to-agent

Demo

Pay per task

Google AP2 demo

Autonomous DeFi

Live

Yield, 24/7, capped

Coinbase Agentic Wallets

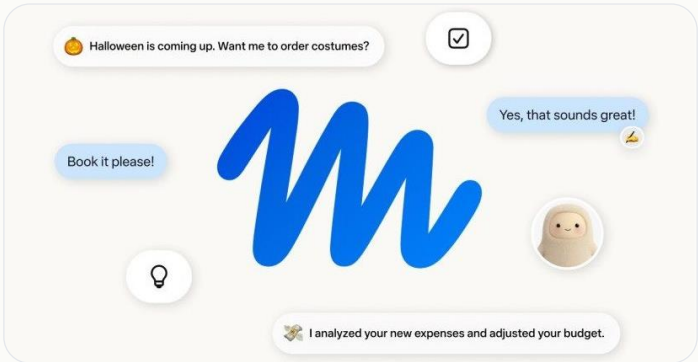
59.5%

of Coinbase agent traders use Grok

Next: custom 21.7%, Claude 13.4% (Brian Armstrong)

What if Meta's Muse had its own wallet?

Today: one-time cards. Idea: a capped USDC wallet.



Muse from Meta, launched 8 Sep 2026 · image: Meta

MUSE WALLET · FUNDED WEEKLY

50.00 USDC

\$5 max each

\$20 max a day

Allowlist

Freeze anytime

MUSE'S DAY, WITH A WALLET

	08:00 · Morning briefing 40 data calls × \$0.01	\$0.40 Paid
	11:00 · Research run 15 min of GPU	\$0.68 Paid
	16:00 · Found a report \$40 report, over limit. Buy?	\$40 Asks you

\$1.08 spent today
8

\$12.01 in card fees, same data

Muse goes shopping: one tap, paid in USDC

M

Muse

200 USDC wallet

Find me trail running shoes, size 10, under \$150

Found one: 129 USDC.

APPROVAL NEEDED

129.00 USDC

Over your \$5 auto-pay cap

Approve

Decline

✓ Paid on Base · receipt saved

- 1

You set a budget
- 2

Muse finds USDC stores
- 3

Over \$5? It asks you
- 4

You approve, it pays
- 5

Store paid, order ships

✓

Real today:

Shopify takes USDC on Base.

⚠

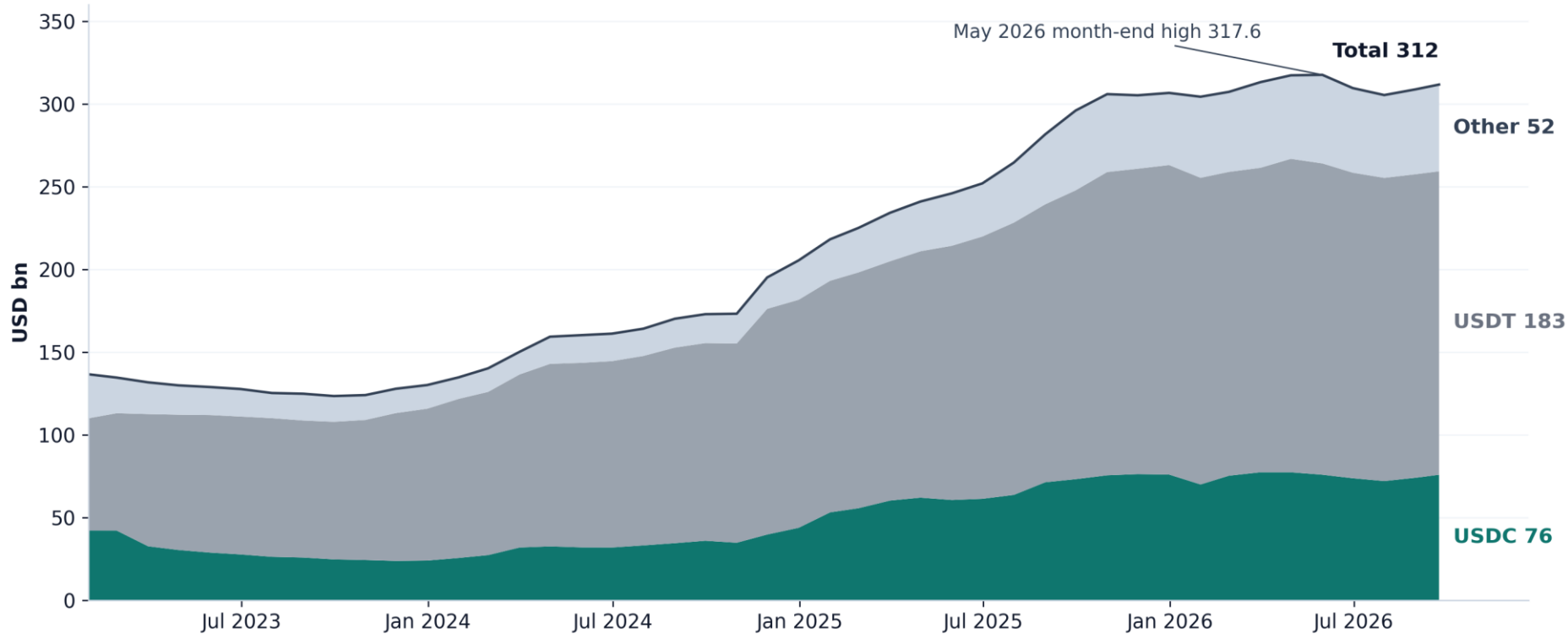
Illustrative:

store, shoe, price. No chargebacks.

ARC RESEARCH | CRYPTO X AGENTS | THE MONEY SUPPLY

Stablecoins more than doubled to \$312bn since 2023, then stalled in 2026

USDC, the coin agent-payment rails default to, is \$76bn, about a quarter of supply; supply has been flat since the spring

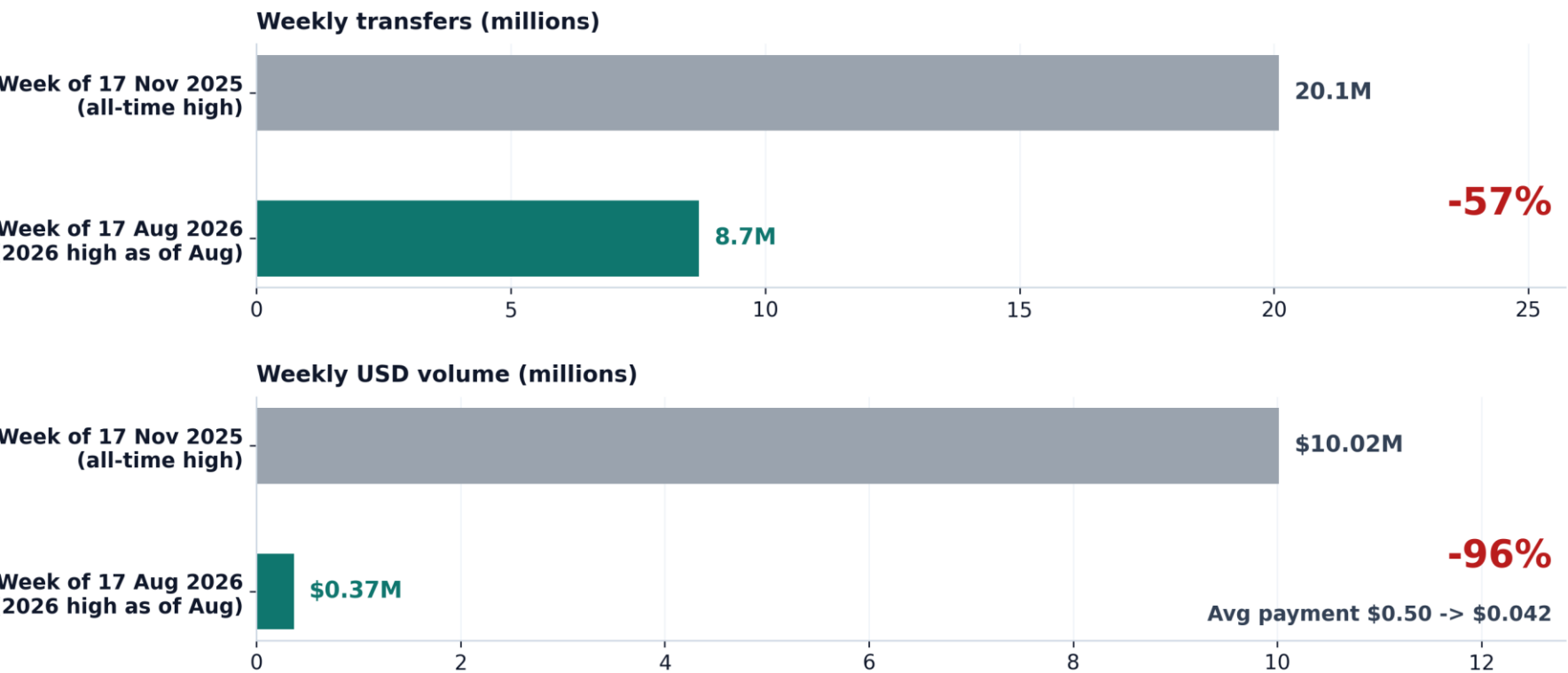


Source: DeFiLlama stablecoins API (stablecoins.llama.fi), all chains, circulating USD-pegged supply; last available day of each month, latest 23 Sep 2026. 'Other' = total less USDT less USDC (derived, our calculation). Includes USDS/DAI, USDe, PYUSD, FDUSD and others. Supply is not payment volume.

ARC RESEARCH | CRYPTO X AGENTS | COUNTS VS DOLLARS

x402 transfer counts rebounded in August; the dollars behind them did not

Peak week vs the busiest week of 2026: transfers down 57%, dollar volume down 96%, so the average payment is now about 4 cents



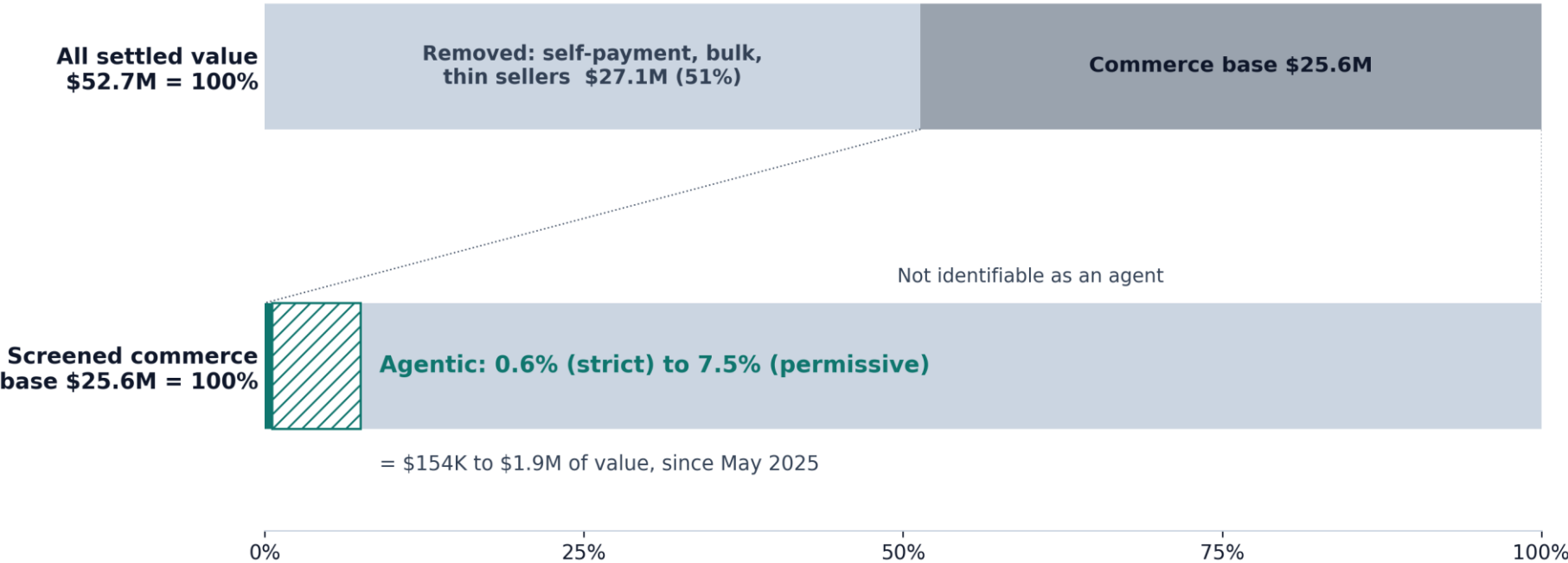
Source: Token Terminal weekly x402 stablecoin transfers and USD volume, as reported by The Cryptonomist and danielmcglynn.com (6 Sep 2026); all chains.
Derived: average payment = weekly USD volume / weekly transfers; % changes our calculation. A weekly time series was not publicly retrievable, so only these two reported weeks are shown.
Separate basis, not plotted: x402scan.com past-30-day totals on 23 Sep 2026 were 22.04M transactions and \$1.01M volume.

03 · ADOPTION DATA

ARC RESEARCH | CRYPTO X AGENTS | HOW MUCH IS REALLY AGENTS?

Of \$52.7M ever settled on x402, AI agents account for roughly \$0.15M-1.9M

Half the flow is self-payment or bulk transfers; of what remains, TRM finds only 0.6%-7.5% looks agentic, about \$5-11K a month



Source: TRM Labs, 'Who's actually paying? Measuring AI agent payments onchain' (9 Sep 2026): \$52.68M settled across 198.9M transactions since May 2025, 99.6% in USDC; \$25.62M after removing self-payments, bulk flows and thin-traffic sellers. Strict test (0.6%) needs sustained multi-month activity plus agent-registry enrollment or multiple sellers; permissive test (7.5%) counts any plausibly agentic address. Derived: removed = 52.68 - 25.62; agentic \$ = share x \$25.62M (our calculation). TRM does not pick between the two tests.

Headline x402 numbers span four orders of magnitude

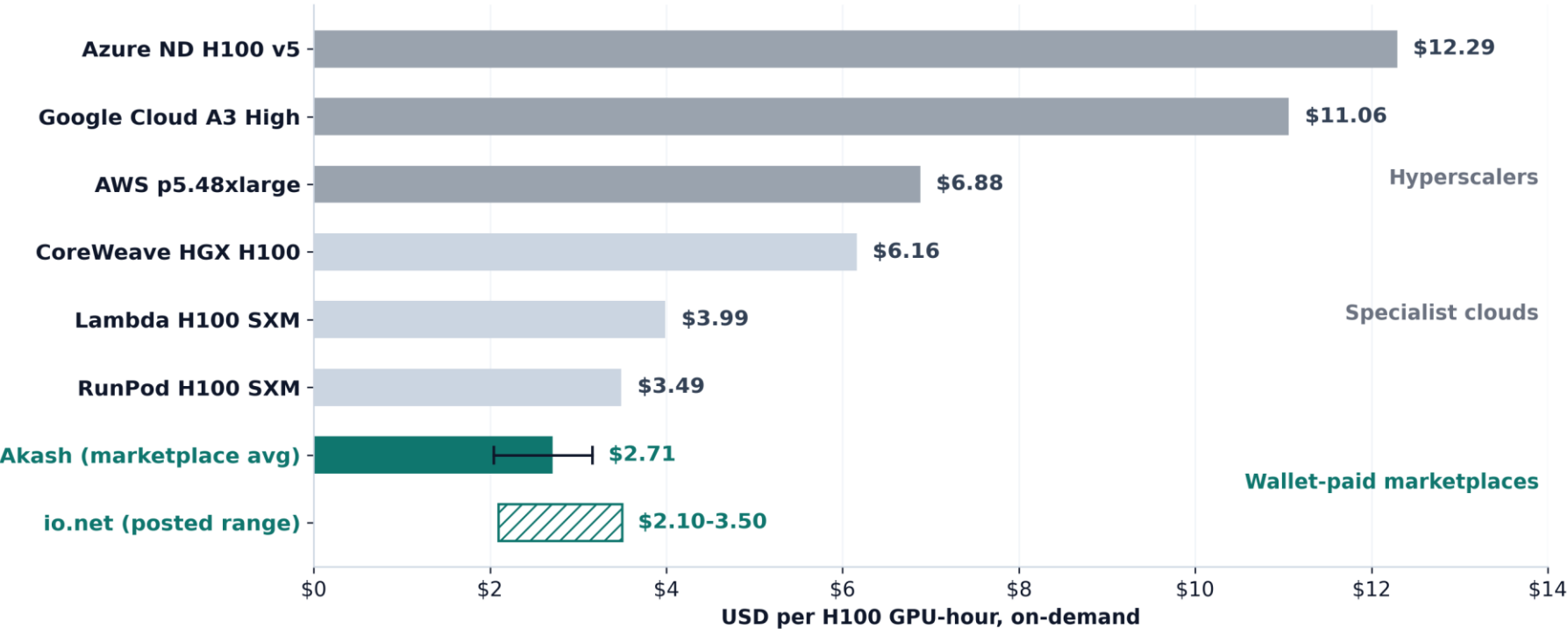
Claim	Measures	Source
\$40–50bn	Unclear	cryptonews.com
~\$600M a year	Run-rate	BlockEden
\$135.7M → \$15.0M	Total; 89% screened out	Artemis & Visa
\$52.7M settled	Total since May 2025	TRM Labs
\$24.2M, last 30 days	Stale counter	danielmcglynn.com
\$1.01M, past 30 days	88% from one server	x402scan

Different measures. The screened ones are small.

ARC RESEARCH | CRYPTO X AGENTS | AGENTS BUYING COMPUTE

Crypto-paid GPU marketplaces post H100 hours at a fraction of hyperscaler list prices

On-demand USD per H100 GPU-hour; the marketplaces an agent can rent from with a wallet sit at \$2-3.50, hyperscalers at \$7-12



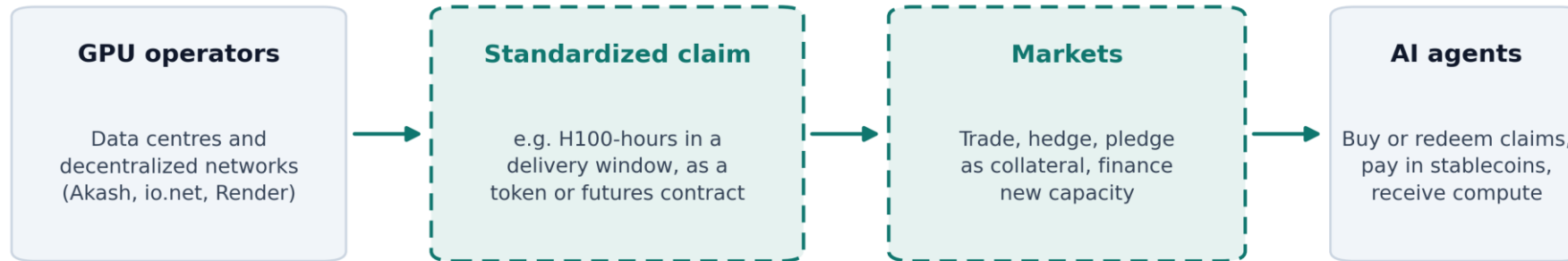
Source: Azure and AWS list prices via instances.vantage.sh; Google Cloud accelerator pricing page; coreweave.com/pricing; lambda.ai; runpod.io/pricing (all checked 23 Sep 2026, US regions, on-demand). Akash: weighted marketplace average and range, akash.network 'The Bid' (12 Sep 2026). io.net: posted range, [io.net](https://io.net/pricing) pricing guide (data labelled April 2026). Nosana lists \$1.50 but showed 0 available hosts on 23 Sep 2026; omitted. Derived: 8-GPU instance prices / 8 (AWS, GCP, Azure, CoreWeave), our calculation. List prices exclude committed-use and spot discounts, which narrow the gap. SXM vs PCIe and interconnect differ.

04 • AGENTS BUYING COMPUTE

ARC RESEARCH | CRYPTO X AGENTS | TOKENIZED COMPUTE CLAIMS

BlackRock's proposal: turn GPU capacity into a standardized, tradable claim

A proposed asset class, not a live market: the paper itself says compute-market liquidity is still limited



Why it would matter for this thesis

- Gives agents a unit to budget and settle compute in, alongside stablecoins for everything else
- Lets operators finance GPUs against future usage, precedent: today's GPU-backed financings
- Open problems the paper flags: chips differ in productivity and energy costs differ by region, so 'one GPU-hour' is not yet a standard unit

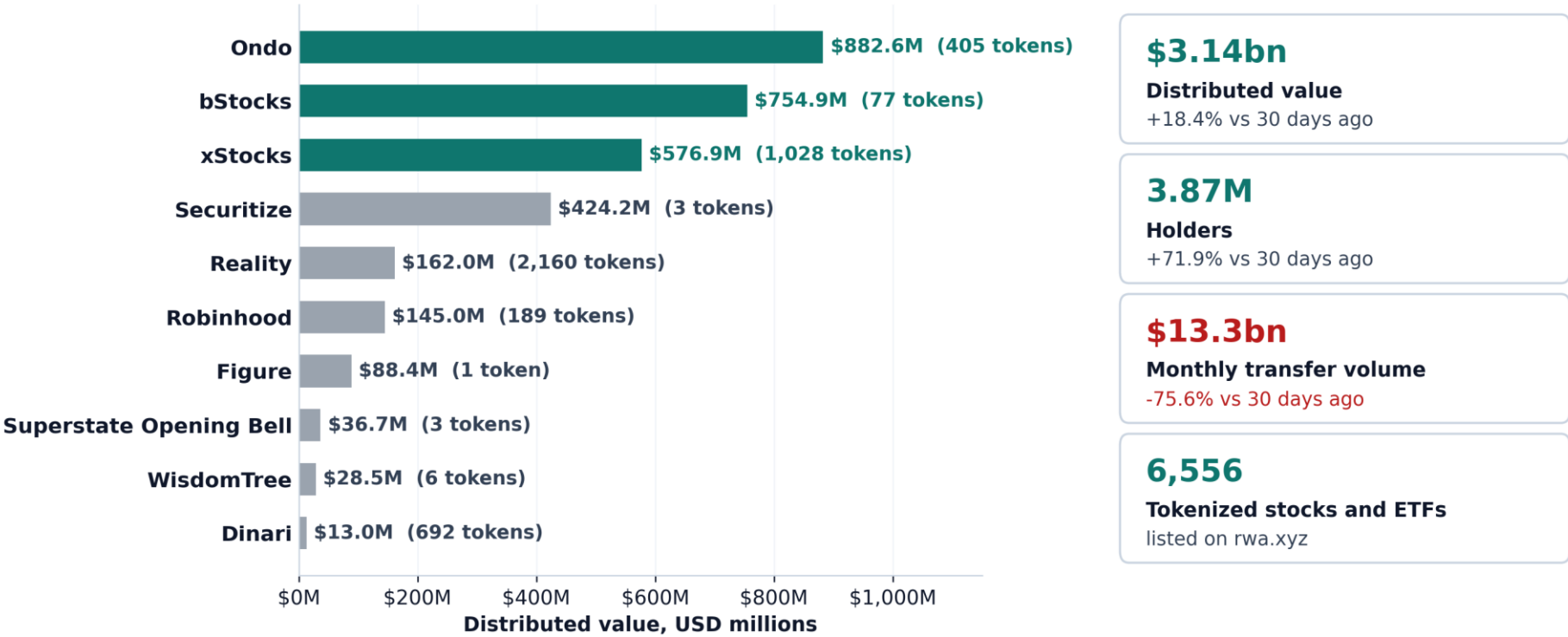
Source: BlackRock, 'The Machine-Native Economy: How digital assets connect intelligence, commerce, and compute' (Sep 2026), pp. 5-10: claims on compute capacity that can be represented, transferred, pledged as collateral and settled onchain; standardized compute contracts and exchange-traded compute futures; existing GPU-backed financings as precedents. Stated obstacles: chip productivity, regional energy cost. Schematic, our illustration of the concept; the paper names no issuer, token standard or venue. Grey boxes = what exists today; teal = the proposed layer.



ARC RESEARCH | CRYPTO X AGENTS | TOKENIZED STOCKS

Tokenized stocks are a \$3.1bn market, and three platforms hold 70% of it

Holders jumped 72% in 30 days to 3.9M, but monthly transfer volume fell 76%: more owners, less trading



Source: rwa.xyz Tokenized Stocks dashboard, 23 Sep 2026: distributed value by platform (tokens held in onchain wallets); represented value (tracked but not distributed) is a further \$27.1M. Counts are the number of tokenized assets per platform. Derived: top-three share = $(882.6 + 754.9 + 576.9) / 3,140 = 70.5\%$; top ten = \$3.11bn of \$3.14bn (our calculation). Includes issuer-sponsored shares and third-party wrapped or synthetic tokens, which may not carry shareholder rights (SEC staff statement, 28 Jan 2026).

Two routes onchain for stocks, and where agents fit

Wall Street rail

- SEC-approved on Nasdaq (Mar 2026)
- DTCC launch in October
- Full shareholder rights

Crypto-native rail

- Ondo, xStocks: 24/7, mostly non-US
- May carry no voting rights
- 3.87M holders, +72% in 30 days

Why it matters for agents

- USDC buys stocks at any hour
- Robinhood agents: since May 2026
- Coinbase agents: since Sep 2026

Six things an agent could do with tokenized stocks



React on a Saturday

Trade the news while Wall Street sleeps

Ondo: 24/7 since Jun 2026



One wallet for crypto and stocks stocks

ETH to NVDA in one swap

xStocks on Solana



Borrow against shares

Borrow USDC; top up before a margin call

Kamino, since 2025



Buy a little, often

\$5 a day, fractional shares



Rules it can't break

Stop-losses and limits you set



Access from abroad

US stocks, no US broker

Non-US users only



Reality check: all possible today; none measured at scale.

Agents already trade on Coinbase; it won't say how much

HOW COINBASE GOT HERE

- Nov 2024
AgentKit
- 11 Feb 2026
Agentic Wallets
- 11 Jun 2026
Coinbase for Agents
- 22 Sep 2026
US stocks and ETFs added

WHAT AN AGENT CAN DO

- Trade 900+ crypto pairs
- US futures and stocks
- Pay for data per call

GUARDRAILS

- 5 USDC cap
- Ring-fenced funds
- You authorize trades
- Revoke anytime

Via ChatGPT, Grok, Claude, Muse

WHICH AI TRADES

Share of agent traders

Grok	59.5%
Custom models	21.7%
Claude	13.4%
Perplexity	3.5%
ChatGPT	1.5%

 **The missing number:** no agent counts, no dollar volume.

Stocks are moving to 23 hours a day, the agent's hours

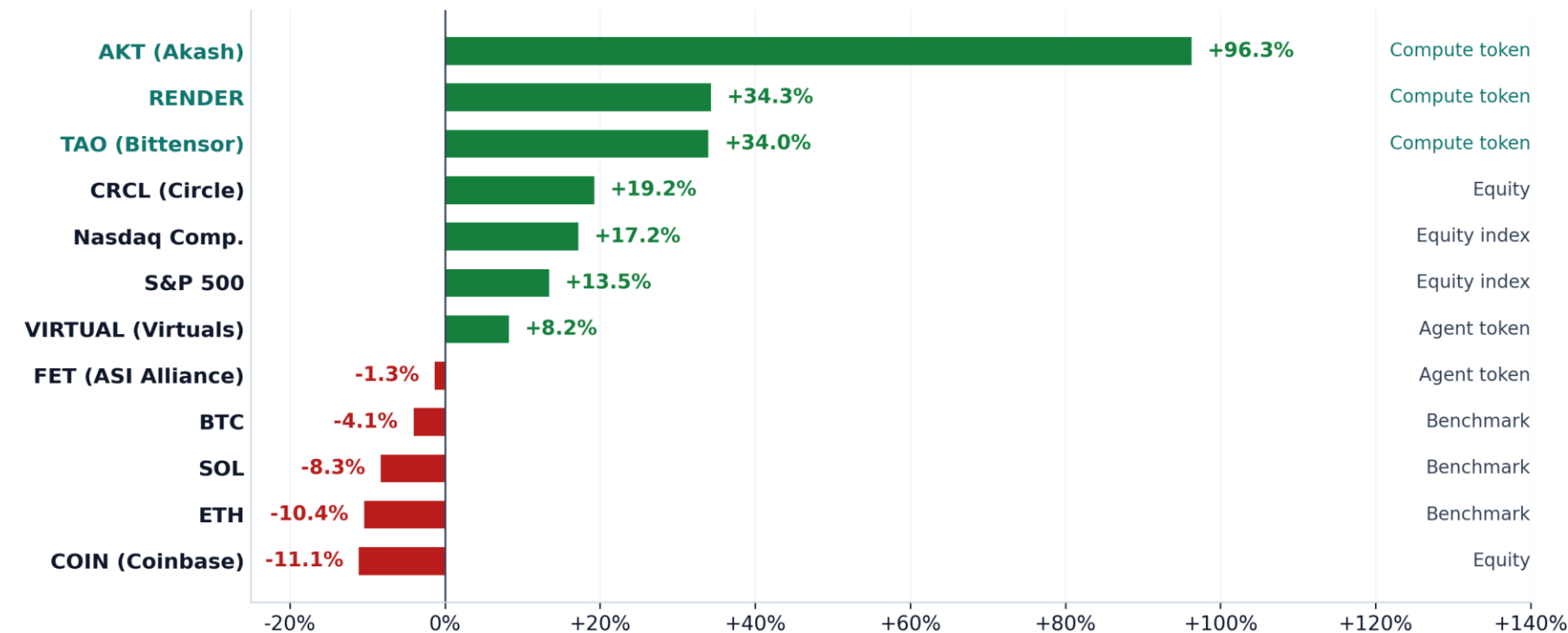


Sources: SEC; Bitcoin.com; The Cryptonomist

ARC RESEARCH | CRYPTO X AGENTS | MARKET SCORECARD

Compute tokens have outrun both Bitcoin and equities in 2026; agent tokens have not

YTD % change: AKT nearly doubled while BTC and ETH fell; the stablecoin issuer (CRCL) beat the exchange (COIN) by 30 points

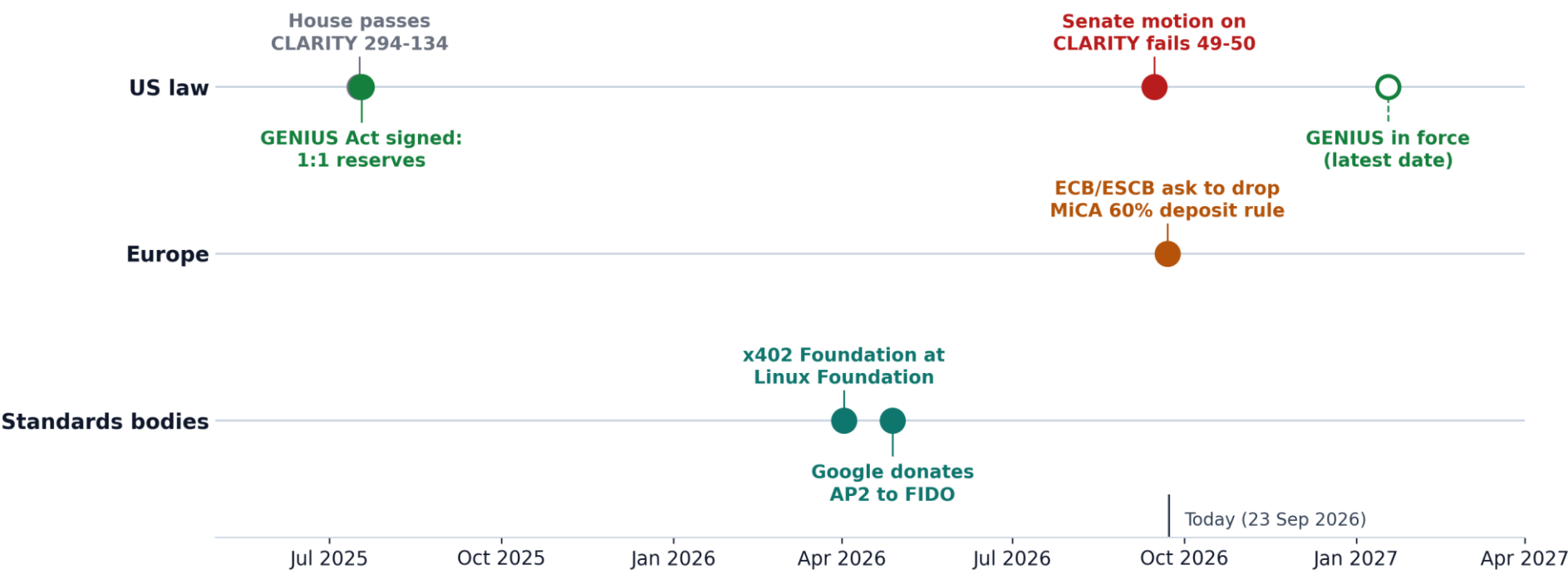


Source: CoinGecko API, 31 Dec 2025 close (00:00 UTC 1 Jan 2026 snapshot) to 23 Sep 2026 17:38 UTC spot. Yahoo Finance 31 Dec 2025 closes; StatMuse 22 Sep 2026 closes for COIN, CRCL, S&P 500, Nasdaq Composite. Derived: % change = latest / 31 Dec - 1, our calculation; price return only, no dividends or staking yield. Crypto and equity end points differ by one day. Small-cap tokens are volatile and illiquid.

ARC RESEARCH | CRYPTO X AGENTS | REGULATION AND STANDARDS

Stablecoin law is settled in the US; market-structure law just stalled

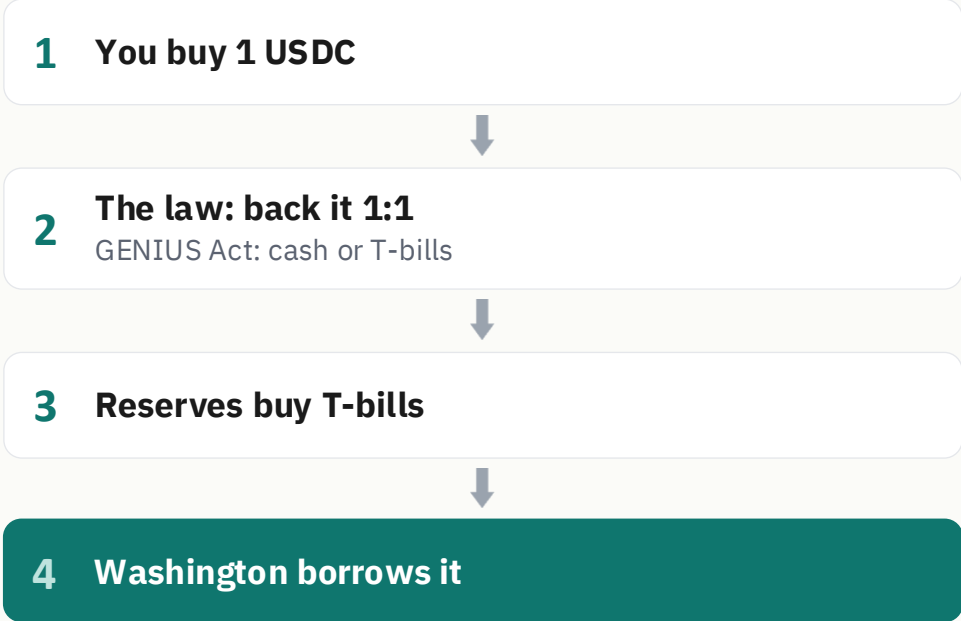
GENIUS takes effect by January 2027 at the latest; CLARITY failed 49-50 on 15 Sep, leaving market-structure rules to agency action



Source: GENIUS Act signed 18 Jul 2025, effective the earlier of 18 months after enactment or 120 days after final rules (congress.gov S.1582; Paul Hastings summary); House Clerk roll call 199, 17 Jul 2025; Jones Day and NPR, 15 Sep 2026; FinTech Weekly and Genfinity, 22-23 Sep 2026 (ESCB response to the MiCA review); Linux Foundation, 2 Apr 2026; Google, 28 Apr 2026. Dashed marker = statutory deadline, not yet reached.

Stablecoins are becoming buyers of US government debt

FROM YOUR DOLLAR TO THE TREASURY



\$120bn+
T-bills held
TBAC, 2025

\$127bn
Tether's Treasuries
Tether, Jun 2025

~\$35bn
Bought in 2025
BIS working paper

–4bp
Bill yield per \$3.5bn inflow
BIS working paper

\$2tn
Market by 2028 (Bessent)
Press-reported

~\$900bn
Extra T-bill demand
Standard Chartered

 **The catch:** a new buyer, but deposits may leave banks.

Own the rails, watch the dollars, not the counts

Would confirm demand

- x402 dollars rise with counts
- Agent share climbs above 0.6%
- Supply tops \$317.6bn
- GENIUS rules final on time

Would break the thesis

- Counts up, dollars flat
- Still 88% from one server
- Agents pick card rails
- Compute tokens outrun usage

Stance: rails and compute over agent tokens, until dollars grow.



SOURCES

Where every number comes from

- DeFiLlama stablecoins API (stablecoins.llama.fi): supply by issuer, pulled 23 Sep 2026
- CoinGecko API: 31 Dec 2025 and 23 Sep 2026 prices; Yahoo Finance and StatMuse: COIN, CRCL, S&P 500, Nasdaq
- TRM Labs, 'Who's actually paying? Measuring AI agent payments onchain', 9 Sep 2026
- Token Terminal x402 weekly data via The Cryptonomist / danielmcglynn.com, 6 Sep 2026; x402scan.com, 23 Sep 2026
- BlackRock, 'The Machine-Native Economy', Sep 2026
- Circle transparency page (circle.com/transparency, checked 26 Sep 2026): USDC reserves and attestations
- Stripe pricing; Coinbase CDP x402 docs; Linux Foundation (2 Apr 2026); Stripe blog (18 Mar 2026); Google blog (28 Apr 2026); Visa IR (14 Oct 2025); Mastercard (29 Apr 2025)
- GPU prices: Google Cloud, CoreWeave, Lambda, RunPod pricing pages; instances.vantage.sh (AWS, Azure); akash.network 'The Bid' (12 Sep 2026); io.net pricing guide
- GENIUS Act (congress.gov S.1582; Paul Hastings); House Clerk roll call 199 (2025); Jones Day and NPR (15 Sep 2026); FinTech Weekly, Genfinity (22–23 Sep 2026)
- Coinbase: Agentic Wallets launch (11 Feb 2026); x402 docs; Coinbase and AWS (15 Jun 2026); Coinbase and Cloudflare (23 Sep 2025); Google AP2 + x402 (16 Sep 2025); CoinDesk (23 Aug 2026)
- Design argument: CoinDesk, Consensus Miami (10 May 2026); The Block, OpenLedger interview (20 Aug 2026)
- Tokenized stocks: rwa.xyz Tokenized Stocks dashboard (23 Sep 2026); SEC staff statement on tokenized securities (28 Jan 2026); SEC approval of Nasdaq rule change (CoinDesk, 18 Mar 2026); DTCC platform (CoinDesk, 4 May 2026); Forbes (19 May 2026); Robinhood newsroom (27 May 2026)
- Agent model share on Coinbase: Brian Armstrong (25 Sep 2026, via TheStreet); Coinbase Developer Platform (16 Sep 2026, via CoinGape); Coinbase for Agents: CDP docs (docs.cdp.coinbase.com, checked 26 Sep 2026), launch (CNBC, CoinDesk, TechCrunch, 11 Jun 2026), stocks and ETFs (The Cryptonomist, 22 Sep 2026); TechTimes (17 Jun 2026)
- Shopping example: Shopify, 'Stablecoins on Shopify' (12 Jun 2025), USDC on Base via Coinbase and Stripe
- 23-hour trading: SEC order on SR-Nasdaq-2025-109 (10 Apr 2026, via Troutman Pepper Locke); Bitcoin.com (18 Aug 2026)
- Tokenized-stock use cases: Ondo blog (25 Jun 2026); xStocks FAQ (docs.xstocks.fi); Kamino collateral (The Defiant, Jul 2025). US debt: Treasury TBAC (30 Apr and 30 Jul 2025); Federal Register GENIUS Act rules (2026); BIS Working Paper 1270; Tether Q2 2025 attestation; The Block (Bessent)
- MPP examples: ByteByteGo, 'AI Agents Can Think. Now They Can Pay.' (Sep 2026)
- Muse: Meta, 'Introducing Muse' (about.fb.com, 8 Sep 2026), text and press image
- Also cited: CoinDesk (11 Mar 2026); BlockEden (5 Mar 2026); cryptonews.com (22 Sep 2026); PayPal Agentic Commerce Pulse (27 Apr 2026)

